2.5 Nature of Light

This topic covers the wave/particle nature of light.

This topic may be studied either by using applications that relate to, for example, solar cells or by the historical study of the nature of light.

Students will be assessed on their ability to:	Suggested experiments
63 explain how the behaviour of light can be described in terms of waves and photons	
64 recall that the absorption of a photon can result in the emission of a photoelectron	Demonstration of discharge of a zinc plate by ultra violet light
65 understand and use the terms threshold frequency and work function and recognise and use the expression $hf = \phi + \frac{1}{2} mv^2_{max}$	
66 use the non-SI unit, the electronvolt (eV) to express small energies	
67 recognise and use the expression $E = hf$ to calculate the highest frequency of radiation that could be emitted in a transition across a known energy band gap or between known energy levels	
68 explain atomic line spectra in terms of transitions between discrete energy levels	Demonstration using gas-filled tubes
69 define and use radiation flux as power per unit area	
70 recognise and use the expression efficiency = [useful energy (or power) output]/[total energy (or power) input]	

Students will be assessed on their ability to:	Suggested experiments
71 explain how wave and photon models have contributed to the understanding of the nature of light	
72 explore how science is used by society to make decisions, for example, the viability of solar cells as a replacement for other energy sources, the uses of remote sensing	